WILO CLOUD MONITORING

Physics-Guided Data Generation & Extraction Formulation Guide

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Technical Specification

1. Physical Reference Model

The system simulates an industrial Wilo centrifugal pump driven by an 8-pole, 3-phase AC induction motor on a 50 Hz European mains supply grid. Baseline parameters are derived from first-principles mechanics and electromagnetic theory.

1.1 Motor and Shaft Rotation

For an 8-pole induction motor on a 50 Hz grid, the synchronous shaft speed is:

$$N_{\text{sync}} = \frac{120 \cdot f_{\text{line}}}{P} = \frac{120 \cdot 50}{8} = 750 \text{ RPM}$$

Synchronous shaft speed — no-load condition

Accounting for slip under rated load ($s \approx 2.67\%$), the nominal shaft speed is **N_{rated} = 730 RPM**. The fundamental shaft rotational frequency is:

$$f_{\text{shaft}} = \frac{N_{\text{rated}}}{60} = \frac{730}{60} \approx 12.17 \text{ Hz}$$

Fundamental shaft rotational frequency

1.2 Defect Frequencies

Blade-Pass Frequency (BPF) — 5-vane centrifugal impeller

$$f_{\text{BPF}} = N_{\text{vanes}} \cdot f_{\text{shaft}} = 5 \times 12.17 = 60.83\ \text{Hz}$$

Blade-pass frequency

Rolling Element Bearing Defect Frequencies — 6-ball bearing, pitch diameter D = 50 mm, ball diameter d = 10 mm, contact angle α = 15°:

Ball Pass Frequency — Outer Race (BPFO)

$$f_{\text{BPFO}} = \frac{N_b}{2} \cdot f_{\text{shaft}} \cdot \left(1 - \frac{d}{D}\cos\alpha\right) = \frac{6}{2} \cdot 12.17 \cdot \left(1 - \frac{10}{50}\cos 15^\circ\right) \approx 29.45\ \text{Hz}$$

Ball Pass Frequency — Inner Race (BPFI)

$$f_{\text{BPFI}} = \frac{N_b}{2} \cdot f_{\text{shaft}} \cdot \left(1 + \frac{d}{D}\cos\alpha\right) = \frac{6}{2} \cdot 12.17 \cdot \left(1 + \frac{10}{50}\cos 15^\circ\right) \approx 43.55\ \text{Hz}$$

Ball Spin Frequency (BSF)

$$f_{\text{BSF}} = \frac{D}{2d} \cdot f_{\text{shaft}} \cdot \left[1 - \left(\frac{d}{D}\cos\alpha\right)^2\right] = \frac{50}{20} \cdot 12.17 \cdot \left[1 - (0.2\cos 15^\circ)^2\right] \approx 29.28\ \text{Hz}$$

Fundamental Train Frequency (FTF)

$$f_{\text{FTF}} = \frac{1}{2} \cdot f_{\text{shaft}} \cdot \left(1 - \frac{d}{D}\cos\alpha\right) \approx 4.91\ \text{Hz}$$

2. Time-Domain Signal Simulation Models

Three physical channels are simulated as discrete time arrays: Vibration Acceleration $a(t)$, Motor Current $I(t)$, and Acoustic Audio $s(t)$. All channels are sampled at $\mathbf{f_s}$ **= 700 Hz** over $\mathbf{T_d}$ **= 2 s**, yielding $\mathbf{N_s}$ **= 1 400 samples** per window.

2.1 Fundamental Physics Models

Damped Impact Model

Shock events (bearing defects, bubble collapse) are modelled as a damped sinusoid. H(t) is the Heaviside step function; ζ = 0.08 is damping ratio; $\omega_n = 2\pi \cdot 1500$ rad/s; $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$:

$$x_{\text{impact}}(t) = A \cdot e^{-\zeta\omega_n(t - t_{\text{imp}})} \cdot \sin\left(\omega_d\left(t - t_{\text{imp}}\right)\right) \cdot H(t - t_{\text{imp}})$$

Damped sinusoidal impulse response

Randomised Periodic Impact Train

Impacts repeat with period $T = 1/f_{\text{fault}}$ and timing jitter δ_i:

$$x_{\text{train}}(t) = \sum_i x_{\text{impact}}^{(i)}(t;\ t_{\text{imp}} = i \cdot T + \delta_i)$$

Periodic impact train with stochastic jitter

Jitter: δ_i ~ U(–0.05T, 0.05T). Amplitude variation: A_i ~ LogNormal(μ, σ).

2.2 Fault-Specific Signal Formulations

Motor Stall

Rotational speed collapses to zero. Current locks at stator-resistance limit (~5.5× rated). Vibration drops to zero as all rotational mechanical force vanishes.

$$I(t) = \sqrt{2} \cdot I_{\text{locked}} \cdot \sin(2\pi \cdot 50 \cdot t) + \text{Noise}, \quad I_{\text{locked}} \approx 5.5 \cdot I_{\text{rated}}$$

Pump Cavitation

Bubble collapse follows the Rayleigh–Plesset equation, producing explosive acoustic bursts. Vibration kurtosis surges. Current exhibits high-frequency load-fluctuation noise.

$$s_{\text{cav}}(t) = A \sum_k \text{damped\_sinusoid}(t,\ t_k) \quad \text{[clustered around BPF rate]}$$

Impeller Damage

Hydraulic unbalance generates periodic force modulations at f_shaft and its harmonics.

$$a_{\text{imp}}(t) = A_d(1 + \sin(2\pi f_{\text{shaft}}\, t)) \cdot \sin(2\pi f_{\text{BPF}}\, t) + \text{Impacts}$$

Shaft Misalignment

Combines angular and parallel unbalance. Sharp spectral peaks appear at 1× and 2× shaft frequency.

$$a_{\text{mis}}(t) = A_1 \sin(2\pi f_{\text{shaft}}\, t + \phi_1) + A_2 \sin(2\pi \cdot 2 f_{\text{shaft}}\, t + \phi_2)$$

Motor Winding Failure

Stator short-circuits introduce 3rd and 5th current harmonics. Partial discharge inter-arrival times follow: p(t) ~ Weibull(λ, k).

$$I(t) = I_1 \sin(2\pi \cdot 50\, t) + I_3 \sin(2\pi \cdot 150\, t) + I_5 \sin(2\pi \cdot 250\, t)$$

Motor Overheating

Winding resistance rises exponentially with temperature. Current efficiency drops; harmonic distortion grows. Audio loudness increases linearly with thermal load.

$$T(t) = T_0 + \Delta T(1 - e^{-t/\tau})$$

3. Statistical Feature Formulation

For signal array $X = \{x_1, x_2, \ldots, x_N\}$ with mean μ and standard deviation σ, the following parameters are computed per sensor window and stored in the monitoring database:

Mean (μ)

Centroid of the data points. Sustained shifts indicate load or fault changes.

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Median

Central value of the sorted sequence. Robust to outliers and impulse spikes.

$$\tilde{x} = \mathrm{percentile}(X,\ 50)$$

Standard Deviation (σ)

Spread of signal energy. Increases under vibration escalation or current fluctuation.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2}$$

Variance (σ^2)

Second statistical moment. Equal to the square of the standard deviation.

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

Range (Peak-to-Peak)

Total waveform span. Highly sensitive to impulsive shock events.

$$\text{Range} = \max(X) - \min(X)$$

Peak

Maximum absolute amplitude — worst-case structural stress indicator.

$$\text{Peak} = \max(|X|)$$

Root Mean Square (RMS)

Total effective signal power. Primary health indicator for vibration monitoring.

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$$

Crest Factor (CF)

Shock indicator — ratio of peak to RMS energy. Healthy bearing: CF ≈ 2.5. Fault onset: CF rises toward 6+.

$$CF = \frac{\text{Peak}}{\text{RMS}} = \frac{\max(|X|)}{\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}}$$

Skewness (S)

Third standardised moment — measures distribution asymmetry. Zero for symmetric signals; positive for right-tailed impulsive faults.

$$S = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^3}{\sigma^3}$$

Excess Kurtosis (K■)

Fourth standardised moment — Fisher definition where K■ = 0 for Gaussian. Most sensitive indicator of impulsive bearing faults; rises sharply above 1.5 at onset.

$$K_e = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^4}{\sigma^4} - 3$$

4. Advanced Diagnostic & Onset Detection Logic

4.1 FFT Fault Frequency Match Score

The FFT transforms each time-domain window to the frequency domain. The top-5 spectral peaks are compared against fault-specific expected frequencies with relative tolerance θ = 10%. The **Fault Frequency Match Score** η is:

$$\eta = \frac{1}{|f_{\text{peaks}}|} \sum_{f \in f_{\text{peaks}}} \mathbf{1}\left(\min_{e \in f_{\text{expected}}} \frac{|f - e|}{e} \leq \theta \right)$$

η ∈ [0, 1] — proportion of top-5 peaks matching known fault frequencies (θ = 10%)

> ■(·) is the indicator function: returns 1 if condition is true, 0 otherwise. A match score $\eta \geq$ **0.6** triggers a fault frequency alert in the database.

4.2 Multi-Sensor CUSUM Onset Detection

Gradual fault onset is identified using a weighted Cumulative Sum (CUSUM) formulation applied to extracted feature vectors across all sensor channels.

Step 1 — Baseline Calibration

Mean μ_{bj} and standard deviation σ_{bj} of feature j are estimated from the healthy baseline window (first 25% of uploaded data).

Step 2 — Z-Score Normalised Deviation

For upload interval i, the normalised deviation of feature j is:

$$z_{ij} = \frac{x_{ij} - \mu_{bj}}{\sigma_{bj}}$$

Z-score relative to healthy baseline statistics

Step 3 — Weighted Deviation Score D■

Fault-specific weights filter direction of change (e.g. current drop on seal leaks, kurtosis rise on bearing wear):

$$D_i = \frac{\sum_j w_j \cdot \max\left(0,\ \mathrm{sgn}_j(z_{ij}) \cdot z_{ij} - z_{\min}\right)}{\sum_j w_j}$$

Weighted deviation score (z_min = 0.8σ sensitivity threshold)

> $\mathrm{sgn}_j \in$ {+1, −1, absolute} matches the physical direction of change for each sensor/feature pair. $z_{\min} = 0.8\sigma$ suppresses noise-level deviations.

Step 4 — CUSUM Accumulation S■

The cumulative score accumulates until a decision threshold is crossed:

$$S_i = \max(0,\ S_{i-1} + D_i - k)$$

CUSUM accumulator with slack parameter k = 0.5

When $\mathbf{S_i}$ **> h = 3.0** for **2 consecutive upload cycles**, the deviation onset index is triggered and an alert event is logged.

4.3 Feature Slope — Rate of Change

For each upload interval i with timestamp t_i (milliseconds), the feature rate of change per second is computed as:

Forward Difference (i < FailureIndex)

$$\mathrm{Slope}_{i,j} = \frac{x_{i+1,j} - x_{i,j}}{(t_{i+1} - t_i)/1000}$$

Forward difference — all intervals before the failure point

Backward Difference (i = FailureIndex, terminal)

$$\mathrm{Slope}_{i,j} = \frac{x_{i,j} - x_{i-1,j}}{(t_i - t_{i-1})/1000}$$

Backward difference — applied at the final (failure) interval only